

MCP Server for HPE Aruba Networking Central

Bridging 2 000+ APIs with AI

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MCP Server for HPE Aruba Networking Central

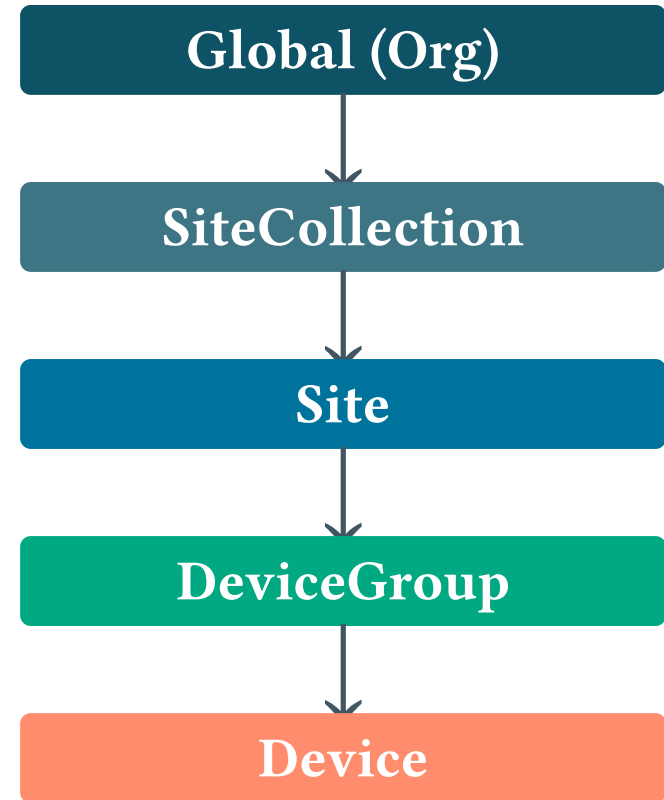
Context

Aruba Central – Configuration Model

Five-level configuration hierarchy

Config propagates **top-down**.
Most specific scope wins.

Precedence (highest → lowest):



Device › DeviceGroup › Site ›
SiteCollection › Global

→ *Deep dive: “Die neue Aruba Central: Praxis-Check der wichtigsten Features”*

An API-First Platform

2 100+ API endpoints

Monitoring,
Configuration,
Alerting,
Firmware,
Topology,

Central + GreenLake

Device
onboarding,
Subscriptions,
Licensing,
Locations, Identity,
...

50 API categories

Full OpenAPI
specs available
on
developer.arubanetworks.com

Troubleshooting,

...

Everything is an API. If it's an API, it's AI-ready.

Design & Architecture

The Challenge

Naïve approach: 1:1 mapping

1 API endpoint = 1 MCP tool

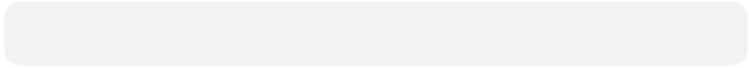
2 121 tools × 8 200 tokens avg



Our approach: Invocation Pattern

3 generic tools + search index

all endpoints accessible



**all
endpoints
accessible**

✓ Fits any context
window

all endpoints accessible

17.4
million
tokens
88 × the
context
window

❌ Impossible

The Invocation Pattern

1. Search

Find endpoints by
keyword

```
unified_search("v
```



2. Discover

Read full API
schema

```
get_api_endpoint_c  
    "GET", "/"  
    config/..."  
)
```



3. Execute

Call the API

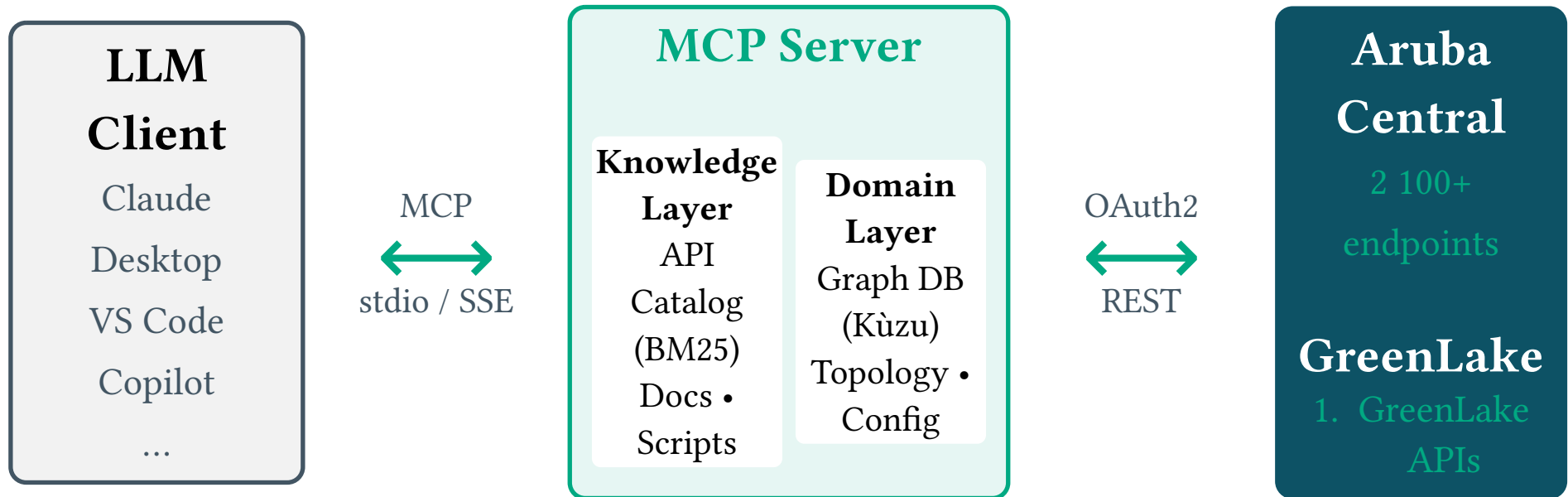
```
call_central_api(  
    "/"  
    config/...", ...  
)
```

all endpoints accessible

The LLM **learns the API on-the-fly** — zero pre-loaded endpoint schemas.

all endpoints accessible

Architecture Overview



all endpoints accessible

Script Engine —
Sandboxed Python
execution

*Seeds auto-populate the graph at startup from live API data.
BM25 full-text search index over all OpenAPI specifications.*

Scripts: Multi-Step Workflows

When to use scripts:

- Multiple API calls in sequence
- Pagination over large datasets
- Complex logic / conditionals
- Reusable automation

Built-in seed scripts:

- populate_base_graph — Org hierarchy

```
from central_helpers import  
api, graph  
  
# Pre-authenticated — no OAuth  
needed  
devices = api.paginate(  
    "network-monitoring/  
v1alpha1/devices"  
)  
  
for d in devices:  
    graph.execute(  

```


all endpoints accessible

- enrich_topology — LLDP adjacency
- populate_monitoring — Ports, clients
- onboard_device — Full onboarding flow

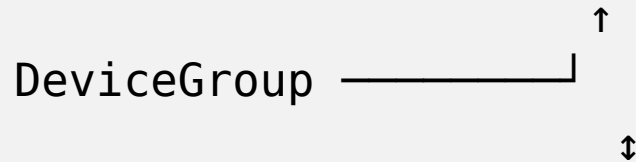
```
"MERGE (d:Device
{serial: $s})"
" ON CREATE SET d.name
= $n",
{"s": d["serial"], "n":
d["name"]}]
)
```

all endpoints accessible

The Graph Database

Embedded graph (Kùzu /
LadybugDB) populated at startup:

```
Org —→ SiteCollection —→ Site  
—→ Device
```



CONNECTED_TO

Example queries:

```
// All devices at a site  
MATCH (s:Site)-  
[:HAS_DEVICE]->(d:Device)  
WHERE s.name = "Berlin-HQ"  
RETURN d.name, d.status
```

```
// LLDP neighbors of a  
switch  
MATCH (d:Device)-  
[c:CONNECTED_TO]->
```

all endpoints accessible

```
adjacency) (LLDP
```

```
(n:Device)  
WHERE d.name = "SW-Core-01"  
RETURN n.name, c.fromPorts
```

- Topology navigation via **Cypher** queries
- LLDP neighbor discovery (incl. unmanaged devices)
- Config scope tracking for blast-radius analysis
- BM25 full-text search across all node types

11 Tools, 4 Categories

☒ Discovery

- `unified_search` — BM25 keyword search
- `list_api_categories` — Browse API areas
- `get_api_endpoint_detail` — Full schema

⚡ Direct API

- `call_central_api` — Central REST calls
- `call_greenlake_api` — GreenLake REST calls

☒ Graph

- `query_graph` — Read-only Cypher queries
- `write_graph` — Enrich graph with findings

☒ Scripts

- `list_scripts` / `get_script_content`
- `save_script` — Persist automation
- `execute_script` — Run in sandbox

Live Demos

Demo Overview



Root Cause Analysis

Find unhealthy devices,



Firmware Compliance

Check firmware versions,



VLAN Configuration

Create VLAN profile,
assign to ports,
verify push status

diagnose problems,
analyze topology

Demo environment

find outdated devices,
recommend upgrades

Demo environment

Lab environment

Demo 1: Root Cause Analysis

“Show me devices with problems in the network and diagnose the root causes.”

Expected agent workflow:

1. Graph query

Query topology for device health

2. Alert API

Fetch active alerts and event details

3. Neighbors

Check LLDP topology for upstream issues

4. Diagnosis

Cross-correlate data and report findings

↑ Replace this slide with a live demo screenshot or video later.

Demo 2: Firmware Compliance

*“Which devices are running outdated
firmware?
Compare against recommended versions.”*

Expected agent workflow:

1. Search APIs

Find firmware
&
compliance
endpoints

2. Get versions

Fetch current
vs.
recommended
firmware

3. Compare

Cross-site
comparison
per device
type

4. Report

Compliance
summary
with
recommendations

↑ Replace this slide with a live demo screenshot or video later.

Demo 3: VLAN Configuration (Mutating)

“Create VLAN 100 on switch [name], assign it to port 1/1/5, ensure trunk ports carry it, and verify the config was pushed.”

Expected agent workflow:

1. Discover

Search config
profile
APIs for
VLANs

2. Create

Create VLAN
config
profile via
API

3. Assign

Assign profile
to
device scope

4. Verify

Check push
status
with
effective=true

↑ Replace this slide with a live demo screenshot or video later.

Closing

Important

⚠ **Not an official HPE product**

This is a **personal / community project** that demonstrates what API-

✓ **Production recommendations**

- Restrict to **GET requests only**
- Full API access = full blast radius

first
platforms enable.

- Use for **monitoring & analytics**
- Be careful with mutations

Try It Out

github.com/tbelz/hpe-networking-central-mcp

☒☒ **Docker**

One-command
setup

☒ **Claude
Desktop**

☒ **VS Code**

Copilot MCP
support

Add to
config.json

Works with any MCP-compatible client • Requires Aruba Central API credentials

Key Takeaway

**An API-first platform is
inherently AI-ready.**

Point an agent at your network —
get autonomous diagnostics, analytics, and
operations.

Closing

Q & A

Questions?

Till Belz

github.com/tbelz/hpe-networking-central-mcp